INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

SHOP EQUIPMENT, ELECTRICAL REPAIR,

SEMITRAILER MOUNTED

This specification is approved for use by the Armament Research, Development and Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mobile, semitrailer-mounted electrical repair shop.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

A-A-1927	-	Padlock
VV-F-800	-	Fuel Oil, Diesel
PPP-B-636	-	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

PPP-B-1055	-	Barrier Material, Waterproofed, Flexible
PPP-T-60	-	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152	-	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture and Fungus-Resistant (For Treatment of Communication, Electronic and Associated Equipment)
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank
MIL-T-704	-	Treatment and Painting of Materiel
MIL-T-740	-	Truck, 5 Ton, 6 X 6, Military Design, M39 Series
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-L-2105	-	Lubricating Oil, Gear, Multi-purpose
MIL-G-10924	-	Grease, Automotive and Artillery
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-G-28554	-	Generator Set, Mobile Electric Power and Supplemental Equipment; Packaging of
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic
MIL-B-46176	-	Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative
MIL-G-52732/11	-	Generator Set, Gasoline Engine-Driven, 10 KW, 60 Hertz and 28 Volt Direct Current, Type I (Tactical) Class 2 (Utility); General Specification for
MIL-G-52889/2	-	Generator Sets, Diesel Engine Driven, 10 KW, 60 Hertz; Tactical, Utility
MIL-V-62038	-	Vehicles, Wheeled: Preparation for Shipment and Storage of
MIL-S-62049	-	Semitrailer, Van: For Installation of Special Equipment, Military Vehicles

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-454	-	Standard General Requirements for Electronic Equipment, Requirement 4
MIL-STD-810	-	Environmental Test Methods
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-1474	-	Noise Level for Army Materiel

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF THE ARMY

DRAWINGS

TA13217E8100 - Shop Equipment, Electrical Repair, Semitrailer Mounted

LIST OF COMPONENTS

SC 4940-95-CL-B05 - Sets, Kits and Outfits, components List, Shop Equipment, Electrical Repair Semitrailer Mounted

(Unless otherwise indicated, copies of the drawings and list of components are available from Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-T, Rock Island, IL 61299-7300.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the document cited in the solicitation (see 6.2.1).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection in accordance with 4.3. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Description. The repair shop shall be in accordance with Top Assembly Drawing TA13217E8100 and

as specified herein. The shop shall include all tools and equipment specified in the Set Components List as compiled from SC 4940-95-CL-B05 (see 6.3).

3.2.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the shop equipment. Any data, such as shop drawings, layouts, flow sheets, processing procedures, prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.3 Initial production. When specified (see 6.2.1), the contractor shall furnish to the Government one or more initial production electrical repair shops for inspection.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5 Shop dimensions and lifting.

3.5.1 Dimensions. Dimensions of the shop shall be as shown on the applicable drawings (see 3.2).

3.6 Body. The contractor shall fabricate and install the equipment on the chassis in accordance with the applicable drawings (see 3.2). The contractor shall furnish all items of equipment including the tools and equipment as specified in 3.2.

3.6.1 Padlock. The contractor shall furnish and install on the rear door and ladder a padlock with hardened shackle and chain in accordance with A-A-1927, type EPA, size 1 1/8 inch. The free end of the chain shall be riveted to the door.

3.7 Electrical system. Electrical system shall be in accordance with the applicable drawings and comply with the NFPA 70.

3.8 Tools and equipment. The tools and equipment, in their as-used condition, shall be stored and anchored to prevent movement during transit. Instruments which do not have transit cases and which are sensitive to damage by vibration shall be protected in their storage compartments by polyurethane closed-cell foam sheet. Tiedown devices shall be designed for a general tool or equipment configuration to allow interchange of like items from different supply sources. Wherever possible, items shall be grouped together by function.

3.9 Safety. All exposed parts, assemblies, or accessories which are subject to high operating temperatures, or which are energized electrically, shall be fully insulated, enclosed, or guarded. All moving parts which are hazardous to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions of the shop.

3.10 Performance.

3.10.1 Installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and tools.

3.10.2 Roadability. When equipped with the tools and equipment specified in SC 4940-95-CL-B05 and loaded in accordance with 3.8, the shop shall withstand towing by a Government loaned M818 truck tractor (see 3.20) on unimproved roads, cross country trails, a chatter course and a diagonal ditch course. During or following towing, there shall be no damage to, or displacement (see 6.8.2) of any component, accessory, part or tool installed in or on the shop and no evidence of defective welded, bolted or riveted connections on the shop.

3.10.3 Rail impact. The shop equipment with all tools and equipment stored in the proper place and with all panels and doors closed, shall withstand a rail impact test in accordance with MIL-STD-810, without damage to the shop or the contents, when impacted at velocities of 4, 6, and 8 miles per hour (mph).

3.11 Environmental conditions.

3.11.1 Heating system. Operate the heaters (see 3.19) continuously for 15 minutes to check its performance.

3.11.2 Air conditioner. Operate the air conditioner (see 3.19) continously for 15 minutes to check its performance.

3.11.3 Temperature (storage and operating).

3.11.3.1 Storage. The shop equipment with all tools and equipment, shall be capable of withstanding storage temperatures down to -50, plus/minus 5, degrees Fahrenheit (F) without encountering any adverse effects or damage. The complete shop shall be capable of operation when returned and stabilized to an ambient temperature within the operating range.

3.11.3.2 Operational temperature(s). The shop equipment shall perform as specified below in any ambient temperature between -25, plus/minus 2, degrees F and +120, plus/minus 5, degrees F:

a. Following a 24 hour soak period at -25, plus/minus 2, degrees F and while at that temperature, operate the installed heater, all portable electric power tools, and electronic equipment from the shop installed power source. Slave starting of the government furnished generator may be attempted if starting difficulties are encountered (see 6.8.8). When returned to ambient temperature, the complete shop shall be operated.

b. Following a 24 hour soak period at +120, plus/minus 5, degrees F and while at that temperature, operate all shop equipment except the installed heater from the shop installed power source.

3.11.4 Sound level. The sound level of the shop shall be in accordance with MIL-STD-1474 Category D

for steady-state noise when operating shop equipment. Hearing protectors need not be supplied with the shop equipment. Hazard signs, if required, shall state: Caution: Hearing Protection Required (see 6.12).

3.12 Reliability. The shop equipment shall demonstrate a mission reliability of at least 0.90 when measured with mean time between failures (MTBF).

3.13 Availability. The shop equipment availability shall be not less than 95 percent.

3.14 Maintainability. The shop equipment shall require a minimum of operational and in-storage maintenance. Replacement components shall be used to the maximum extent possible. The shop equipment shall require a minimum of man-hours and skill to accomplish maintenance. Ease of maintenance and restricted use of components requiring special skills or tools for maintenance shall be stressed. The maintenance-operation ratio shall be no greater than 0.12 (see 6.8.1).

3.15 Fuels, lubricants, and coolants. The shop(s) shall be assembled and tested with the products of the type and grade specified in table I for the applicable ambient air temperature. The shops shall operate as specified herein when furnished with these products (see 6.5).

TABLE I. Service Products

Item	Application	Specification	Type & Grade
1.	Engine:		
	(a) Heavy Duty	MIL-L-2104	10, 30
	(b) Arctic	MIL-L-46167	One Grade
2.	Fuel:		
	Diesel Fuel	VV-F-800	
3.	Grease:		
	(a) Automotive	MIL-G-10924	One Grade
	(b) High Temperature	MIL-G-23827	One Grade
4.	Hydraulic-system Oil:		
	(a) Oil	MIL-L-2104	10
	(b) Oil (low temperature)	MIL-L-46167	One Grade
5.	Lubricating Oils:		
	Universal	MIL-L-2105	80, 90, 140
6.	Brake Fluid	MIL-B-46176	One Grade

3.15.1 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.16 Fungus resistance. Electrical and Electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not weather resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.17 Identification marking and instruction plates.

3.17.1 Electromagnetic interference (EMI). A permanent warning notice, indicating that EMI could result if electric power tools/equipment are operated in close vicinity to electronic communication or surveillance equipment, shall be included and displayed in a conspicuous place(s) on the shop van exterior entrance(s) and on the master power control panel.

3.17.2 Identification marking. The shop shall be identified in accordance with MIL-STD-130.

3.17.3 Instruction plates. Instruction plates shall conform to MIL-P-514, Type III, Composition C, or Type 1, Style 1, Composition A, Class 2.

3.18 Treatment and painting. The portions of the shop normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.19 Government furnished property (GFP). Unless otherwise specified (see 6.2.1), the following property in the quantities indicated shall be furnished by the Government (see 6.7).

Item No.	Description	Identification	Quantity for each shop
1	Semitrailer, Van, Shop, Folding Side, 6-Ton, 4-Wheel, with air conditioner and heaters.	M313 MIL-S-62049 EPL 8736316	1

3.20 Government loaned property (GLP). Unless otherwise specified (see 6.2.1), the following property in the quantities indicated shall be loaned by the Government (see 6.6):

Item No.	Description	Identification	Quantity
1	Truck-tractor, M818 NSN 2320-00-050-8984	MIL-T-740	1
2	Generator Set, Gas Engine 10KW, AC, 120/208V., Three-Phase NSN 6115-00-889-1447	MIL-G-52732 MEP018A	1
	or		
	Generator Set, Diesel Engine 10KW, AC, 120/208V., Three-Phase NSN 6115-00-465-1030	MIL-G-52889 MEP-003A	

3.21 Workmanship. Standards of workmanship shall be such that the shop shall have the safety, stability, and operating characteristics specified herein and shall be free of defects that could adversely affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations or tests) as specified herein. Except as otherwise specified in the contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor shall be responsible for ensuring that components and materials used are manufactured, tested, and inspected in accordance with referenced specifications and standards.

4.1.3 Preparation for examination and tests. Prior to examination and test, the shop shall be lubricated with oils and greases specified in 3.15, Table I. Oils shall be those designated for use in the ambient temperature at which the examinations and tests are being performed.

4.1.4 Examination or test of GFP or GLP items. Failure of GFP (see 3.19) or GLP (see 3.20) during examinations or tests does not constitute failure of the examination or test. This includes failure of GFP or GLP engines to start during the tests. Slave starting may be used. Failure of GFP and GLP items are to be reported through normal channels.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Packaging inspection (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined for the defects marked "X" in Column 3 of Table II. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.3.3.

TABLE II. Examination schedule

Defect	Requirement Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5
Assembly incomplete or incorrect	3.2	X	X	X
Material not as specified	3.4	X	-	X
Dimensions not as specified	3.5	X	-	-
Body and equipment not as specified	3.6 thru 3.7, 3.19	X	X	X
Parts, components, tools or equipment missing or as specified	3.8	X	X	X
Stowage or tiedowns not as specified	3.8	X	X	-
Safety devices and guards missing or not as specified	3.9	X	X	X
Lubrication not as specified.	3.15, 3.15.1	X	---	X
Fungus and moisture resistance not as specified	3.16	X	---	X
Identification marking, Warning and instructions, notices and plates not as specified	3.17	X	X	X
Treatment or painting not as specified	3.18	X	X	X
Workmanship not as specified	3.21	X	X	X

4.3.2 Tests. Each repair shop shall be tested as specified in Column 1 of Table III. Failure of any test shall be cause for performing the inspection specified in 4.3.3.

4.3.3. Disassembly inspection. A first article shop failing a performance test shall be disassembled in the presence of a Government representative to the extent necessary to determine the cause of the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected. Reassembly with accepted parts and retesting shall be the responsibility of the contractor.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production shops shall be selected at random by the Government from shops being produced from production tooling and will be examined for the defects marked X in Column 4 of Table II and subjected to the tests marked X in Column 5 of Table III to determine conformance to the requirements of this specification. The inspection shall be performed by the Government at a site selected by the Government. Acceptance of an initial production shop shall not exclude the remaining shops from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production shop to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on shops previously produced and accepted under the contract. Any deficiencies found as a result of the initial production inspection shall be considered prima facie evidence that all shops accepted prior to completion of initial production inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each shop shall be examined for the defects marked X in Column 5 of Table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Shops shall be subjected to the tests marked X in Column 6 of Table III. Sampling for tests is to be one of each ten shops for power tool operation (4.6.2.2) and one of each twenty shops for paint finish (4.6.2.10).

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table II.

4.6.2 Tests.

4.6.2.1 Test schedule. Tests shall be in accordance with table III. Sequence of testing shall be at the option of the test facility, except as specified in the paragraphs herein.

TABLE III. Test schedule.

Test	Test Paragraph	Requirement Paragraph	First Article	Inital Production	Quality Conformance
1	2	3	4	5	6
Power Tools	4.6.2.2	3.2 and 3.10.1	X	X	X
Roadability	4.6.2.3	3.10.2	---	X	---
Impact Testing	4.6.2.4	3.10.3	X	---	---
Environmental	4.6.2.6	3.11	X	---	---
Sound level	4.6.2.5	3.11.4	X	---	---
Reliability	4.6.2.7	3.12	---	X	---
Availability	4.6.2.8	3.13	---	X	---
Maintainability	4.6.2.9	3.14	---	X	---
Paint Finish	4.6.2.10	3.18	X	---	X

4.6.2.2 Installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and power tools for a period of at least 2 minutes on each power source (see 3.10.1). Inability of installed equipment or power tools to function properly shall constitute failure of this test.

4.6.2.3 Road tests. Load the shop with the tools and equipment specified herein and perform the following tests, per requirements of 3.10.2.

(a) Unimproved roads. Drive the shop 50 miles on unimproved roads at maximum safe speeds. The course shall consist of poorly maintained gravel and dirt roads with rutting, and occasional potholes.

(b) Cross-country trails. Drive the shop 50 miles on cross-country trails at maximum safe speeds. The course shall consist of native soil with rock added as required to maintain passability. Surface conditions shall range from smooth to rough with rutting, potholes, and depressed areas.

(c) Chatter course. Drive the shop over a chatter course until at least 2,600 either upward or downward accelerations of the magnitudes specified herein have been applied to a rear wheel center. The course shall be of such construction that the acceleration magnitudes specified below can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than three cps. The maximum acceleration magnitude specified below shall not be exceeded and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives in one pass over the course at least 50 consecutive vertical acceleration cycles of any magnitudes less than the maximum specified herein. Measure the accelerations by a trial test to establish the required vehicle test speed. Then disconnect the recording instruments and complete the test at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually an axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the acceleration shall be not less than four Gravity (Gs) nor more than seven Gs.

(d) Diagonal ditch course. Drive the shop over a series of ditches until 600 reverse torsional frame stresses have been applied. The ditch profile shall be such so as to allow vertical wheel travel of 18 inches from crest to root. The ditches shall be at 45 degrees to the centerline of the course and spaced a sufficient distance apart to allow only one rear wheel and one front wheel of opposite sides in the ditches at any one time.

Examine the shop at the completion of each phase of the road test. Damage to, or displacement (see 6.8.2) of any component, accessory, part, or tool; evidence of defective welded, bolted, or riveted connections excluding the Government furnished chassis shall constitute failure of these road tests.

4.6.2.4 Rail impact test. Conduct rail impact test in accordance with MIL-STD-810 with the exception of speeds which shall be 4, 6, and 8 miles per hour in one direction and at 8 miles per hour in the reverse direction. Tests are to be conducted with all equipment stored in the proper locations and all panels and doors closed (see 3.10.3).

4.6.2.5 Sound level measurement. Measure the sound level at the work station while operating each of the power tools. Sound level in excess of Category D of MIL-STD-1474 shall require the installation of hazard signs as specified in 3.11.4.

4.6.2.6 Environmental.

4.6.2.6.1 Heating system test. Operate the heaters continuously for not less than 15 minutes to check the operation of the heaters, thermostat, and heat deflectors (see 3.11.1). Any of the following that occurs as a result of faulty material supplied by the supplier or incorrect installation by the supplier shall constitute failure of this test:

a. Malfunctioning of deflector or fuel supply line.
b. Smoking, sooting, or back draft.
c. Entrance of fuel vapors or exhaust gases into the interior of the shop.

4.6.2.6.2 Air conditioner test. Operate the air conditioner continuously for not less than 15 minutes to check the operation of the air conditioner and thermostat (see 3.11.2).

4.6.2.6.3 Temperature (storage and operation) tests. Maintain the shop at an ambient temperature of -50, plus/minus 5, degrees F for a period not less than 72 hours (see 3.11.3.1). At the end of this period raise the temperature to -25, plus/minus 2, degrees F for a period not less than 12 hours. At the end of this time period and with the temperature at -25, plus/minus 2, degrees F, the following operational tests shall be performed (see 3.11.3.2):

a. Inspect the entire unit for any damage which may have resulted from exposure to temperature extreme.
b. Start and operate the heating system. The heater shall operate as required in 4.6.2.7.1.
c. Start and operate individually all portable electric power tools from the shop installed power source for a minimum of 2 minutes.

After the conclusion of the operating test at -25, plus/minus 2, degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +70, plus/minus 5, degrees F. When stabilized at +70, plus/minus 5, degrees F, start and operate individually all installed equipment and power tools from the shop installed power source for a minimum of 2 minutes.

After the conclusion of the operating test at +70, plus/minus 5, degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +120, plus/minus 5, degrees F for a period of 24 hours. After the 24 hours, soak at +120, plus/minus 5, degrees F, start and operate individually all installed equipment and power tools from the shop installed power source for a minimum of 2 minutes. Nonconformance to 3.11 shall be cause for rejection of the unit.

4.6.2.7 Reliability. The initial production shop shall be tested for a total of ten missions to determine conformance to the requirements of 3.12. Data shall be gathered to evaluate the system reliability. Reliability is to be evaluated from MTBF and mission operating time (see 6.8.4 and 6.8.5).

$$\text{Reliability} = e^{-(t/u)}$$

where,

t = mission operating time, hour
u = mean time between failures (MTBF), hour
e = 2.718

The shop shall be considered to have demonstrated the required reliability if zero chargeable reliability failures occur (see 6.8.3).

4.6.2.7.1 Mission operation and test. A mission shall consist of the following:

a. Drive the fully loaded shop set at maximum safe speeds for ten miles on unimproved roads and ten miles on cross-country trails.
b. Using the instructions furnished with the shop, set up the shop for

operation.
c. Operate each installed equipment and power tool for a period not less than 5 minutes one time per mission, except as noted.
d. Prepare the shop for movement to another location, to begin the next mission.

Nonconformance with 3.12 shall constitute failure of this test.

4.6.2.8 Availability. During the performance of the reliability test as specified in 4.6.2.7, data shall be gathered to evaluate the system availability. Availability is to be evaluated with respect to operating time and corrective maintenance (see 6.8.5 and 6.8.6).

$$\text{Availability} = \frac{(\text{MTBUMA})}{(\text{MTBUMA} + \text{MTTR})} (100 \text{ percent})$$

where,

MTBUMA = Mean time between unscheduled maintenance actions (unscheduled means corrective maintenance actions).

MTTR = Mean time to repair (use only times associated with corrective maintenance) (see 6.8.7).

The system availability requirement shall conform to 3.13. Nonconformance with 3.13 shall constitute failure to meet the availability requirement.

4.6.2.9 Maintainability. Maintenance operations required during first article and initial production tests shall be gathered to evaluate the system maintainability. Maintainability is to be evaluated with respect to total maintenance man-hours and total operating time during testing.

$$\text{Maintenance Ratio} = \frac{\text{TNMMT}}{\text{TOT}}$$

where,

TNMMT = Total number of man-hours of maintenance (scheduled maintenance and unscheduled repairs) during testing.

TOT = Total operating time (see 6.8.5).

Any support elements, i.e., technical publications, repair parts, special tools, etc., provided under the contract for assisting or conducting maintenance shall be evaluated. The system maintenance ratio shall conform to 3.14. Nonconformance with 3.14 shall constitute failure of this test.

4.6.2.10 Paint finish test. The hardness adhesion and thickness of the paint used on the vehicle shall be determined (see 3.18) as follows:

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the vehicle until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.
b. A test sample of the same metal, painted with the same process and at the same time as the vehicle, shall have a sharpened H lead pencil pressed into the paint as on "a". Failure of the surface

to pit (paint too hard) shall be cause for rejection.
c. Two two-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.
d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least the thickness specified on the applicable drawings shall be cause for rejection.

4.7 Inspection comparison. The Government may select shops at any time during the contract production period and subject these shops to the examination specified in Column 3 of Table II and the tests specified in Column 2 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Shops will be selected at random from those which have been accepted by the Government and shops will not include the previously inspected first article model or initial production model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison shop to meet any requirement specified herein during and as a result of the examination and test specified in 4.7 shall be cause for rejection of the inspection comparison shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on shops previously produced and accepted under the contract. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all shops accepted after approval of the initial production shop are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Packaging inspection. Packaging inspection shall be conducted to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Level C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with method C-1 of MIL-P-116, except that no class 1 ozone depleting substances such as 1.1.1 trichloroethane, halons, chlorofluorocarbons or carbon tetrachloride shall be used.

5.1.1.2 Drying. Drying shall be accomplished by any applicable means stated in MIL-P-116.

5.1.1.3 Preservative application. Immediately after cleaning and drying surfaces of any item in the shop set that are subject to corrosion shall be treated with a P-type preservative in accordance with MIL-P-116, except that preservative P-6 shall not be used. Excessive use of preservative compounds shall be avoided. Petroleum (P-type) preservative compounds shall not be applied to non-metallic parts, corrosion resistant metal or metal that is treated or painted to resist corrosion.

5.1.1.4 Sets or kits. Items furnished in sets or kits with a carrying case shall be preserved as specified in 5.1.1.3, replaced in their case and placed in their proper storage compartment.

5.1.1.5 Items having mounting brackets or holders. Items furnished with mounting brackets or holders shall be preserved as specified in 5.1.1.3 and replaced in the bracket or holder and secured to prevent movement.

5.1.1.6 Handtools and tool sets. Handtools and tool sets shall be preserved in accordance with 5.1.1.3 and placed in their proper storage compartment.

5.1.1.7 Soldering equipment. Soldering equipment shall be preserved in accordance with 5.1.1.3. Those items that come in a carrying case shall be preserved as specified in 5.1.1.3, replaced in their case and placed in their proper storage compartment.

5.1.1.8 Hardware. Hardware such as bolts, nuts, screws and washers shall be preserved in accordance with 5.1.1.3 and placed in their proper storage compartment.

5.1.1.9 Items furnished in commercial containers. Items and materials such as lubricating oils, silicon compound, and grease, furnished in sealed cans, pails, drums, jars or tubes from the manufacturer do not require additional preservation or packing, and shall be placed in their proper storage compartment.

5.1.1.10 Overall storage. To prevent any type of physical or functional damage all items in the shop set shall be placed in their proper storage compartments, carrying cases, mounting brackets, shelves, drawers or boxes and anchored, blocked, braced and cushioned in accordance with MIL-STD-1186.

5.1.1.11 Electrical wire. The ends of electrical wire shall be sealed with tape to prevent moisture from contacting wire. Loose wire shall be coiled to a safe minimum diameter and secured in three equidistant places with cotton twine or tape. Wire on reels shall have both ends taped if possible otherwise just the exposed end shall be taped, the reel shall be d to r

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, H-1, or H-5, secured in place with tape of PPP-T-60, type IV.

5.1.1.13 Air conditioner. Exterior body openings shall be sealed with tape c

5.1.1.14 Locks and keys. The interior of locks shall be coated with a dry lubricant and the locks operated to assure penetration of the lubricant. The locks used for securing the doors and doorsills for shipment and shall be protected by a wrap of barrier material conforming to MIL-B-121, type I, grade A. The wrap shall be secured with tape conforming to PPP-T-60, type IV. A cloth or paper shipping tag containing the shop equipment identification number and the trailer registration number shall be attached to the related padlock keys f

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) First article when required (see 3.1).
(d) When initial production inspection is required and number of shops to be furnished (see 3.3).
(e) Government-furnished property if different (see 3.19).
(f) Government-loaned property if different (see 3.20).
(g) Preservation as required (see 5.1).

6.3 Set components list. The contracting officer should arrange to furnish information concerning items specified in Set SC-4940-95-CL-B05 (see 3.2).

6.4 Technical interface data. The contracting officer should arrange to provide information concerning technical interface for the Government furnished semitrailer specified in 3.19.

6.5 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the shop covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.15).

6.6 Government-loaned property. The contracting officer should arrange to loan the contractor the property specified in 3.20.

6.7 Government-furnished property. The contracting officer should arrange to furnish the supplier the property specified in 3.19.

6.8 Definitions.

6.8.1 Maintenance ratio. The ratio of the number of maintenance man-hours required for both scheduled maintenance and unscheduled repairs to the number of hours of operation (see 3.14).

6.8.2 Displacement. Displacement is defined as dislodgment of any tool or item of equipment from its designated position. Movement of a tool or item of equipment within its designated position without any dislodgment is permitted (see 3.10.2 and 3.14).

6.8.3 Chargeable reliability failure. A chargeable reliability failure is defined as any malfunction:

a. Which cannot be corrected by organizational repair or replacement action using the controls on equipment tools or parts.
b. Which causes or may cause:
    (1) Failure to start operation.
    (2) Cessation of operation.
    (3) Degradation of performance below specified levels.
    (4) Damage to the shop by continued operation.
    (5) Safety hazards to personnel.

6.8.4 Mean-time-between-failure (MTBF). For a particular interval, the total functioning life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, cycle, miles, events, or other measure of life units (see 4.6.2.7).

6.8.5 Operating time (OT). Defined to be an 8-hour period per mission. This time does not include scheduled maintenance or unscheduled repair time. If the mission profile is completed in less than 8 hours, the balance is considered to be standby time (see 4.6.2.7 and 4.6.2.9).

6.8.6 Corrective maintenance action. Any test incident that requires unscheduled maintenance time and includes both reliability chargeable failures and nonchargeable incidents (see 4.6.2.8).

6.8.7 Mean time to repair (MTTR). The total corrective maintenance time in clock hours divided by the total number of corrective maintenance actions (see 4.6.2.8).

6.8.8 Slave starting. External electrical power applied to a generator whose batteries have become discharged which may receive electrical battery voltage from an outside source through a slave cable plugged into the slave receptacle.

6.9 Packing (see 5.2). Levels of packing have not been included for semitrailer and mounted equipment as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing markings, the level of packing should be considered level A and should be marked A/A.

6.10 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.11 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of shops.

6.12 Ear protection. When the noise level exceeds category D of MIL-STD-1474, the operator's manual for the shop shall include appropriate instructions for use of hearing protection when using equipment that produces noise above 85 decibels (dB) (A) inside and outside the shop, and within a distance from the shop at which 85 dB(A) will be met.

6.13 Subject term (key word) listing.

Environmental	Semitrailer	Tools and equipment
Roadability	Maintainability	Reliability

6.14 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Army - AL

Preparing activity:
Army - AL

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